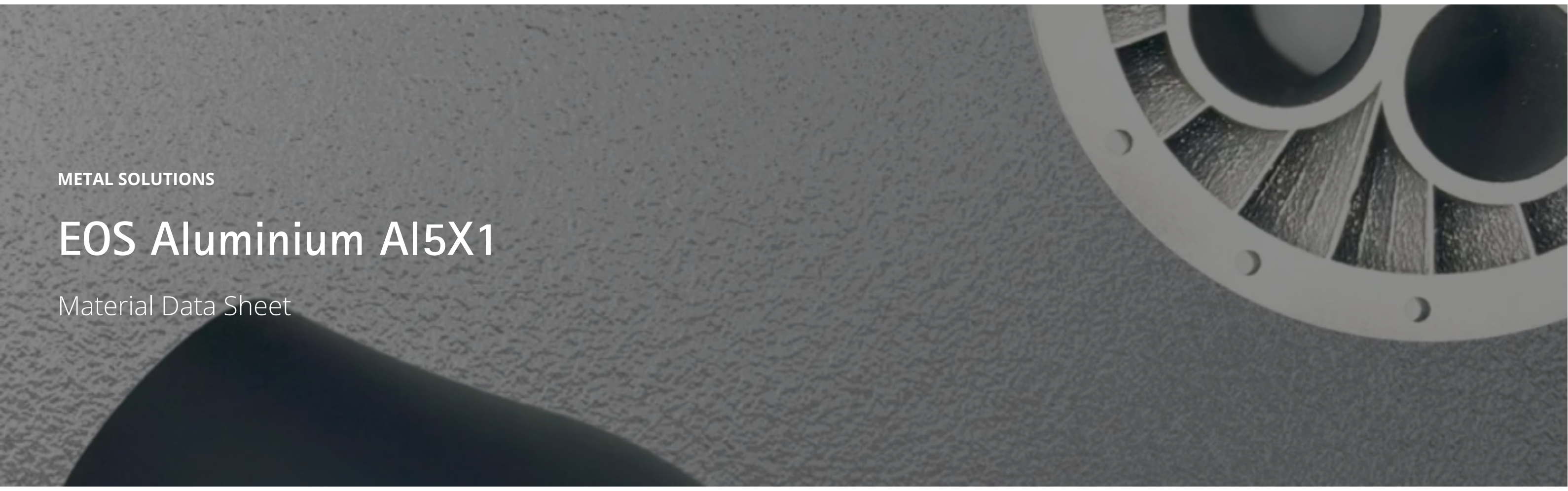
Status as of 18.08.2026



METAL SOLUTIONS

EOS Aluminium Al5X1

Material Data Sheet

EOS ALUMINIUM AL5X1

High Strength & High Elongation Aluminium for AM

EOS Aluminium Al5X1 is a heat-treatable aluminium alloy designed for AM to offer a compelling combination of high strength and high elongation. Al5X1 exhibits excellent mechanical properties with a strength above 400 MPa and an elongation exceeding 13% after heat treatment. The recommended single-step heat treatment does not require a water quench and enables robust part production.

MAIN CHARACTERISTICS

- Excellent combination of strength & elongation
- Good corrosion resistance
- Parts can be anodized

TYPICAL APPLICATIONS

- Aerospace
- Automotive
- Marine
- Lightweight designs

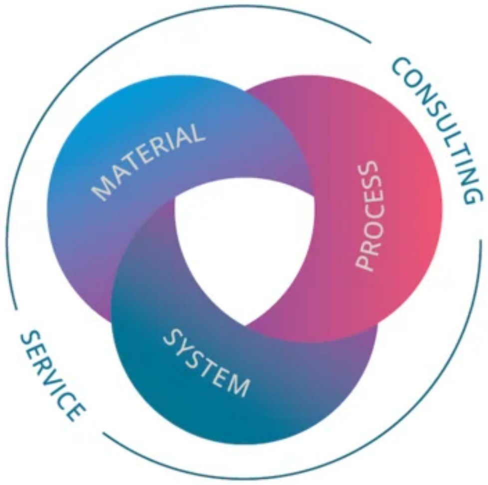
The EOS Quality Triangle

EOS uses an approach that is unique in the AM industry, taking each of the three central technical elements of the production process into account: the system, the material and the process. The data resulting from each combination is assigned a Technology Readiness Level (TRL) which makes the expected performance and production capability of the solution transparent.

EOS incorporates these TRLs into the following two categories:

- → Premium products (TRL 7-9): offer highly validated data, proven capability and reproducible part properties.
- → Core products (TRL 3 and 5): enable early customer access to newest technology still under development and are therefore less mature with less data.

All of the data stated in this material data sheet is produced according to EOS Quality Management System and international standards



POWDER PROPERTIES

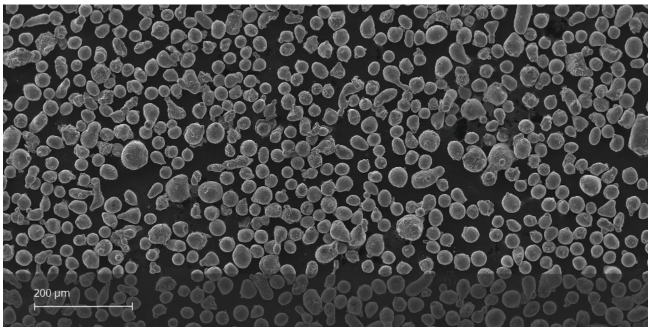
The chemical composition of EOS Aluminium Al5X1 is specially designed for AM. Powder composition values shown below are typical.

Powder Chemical Composition (wt.-%)

Element	Min.	Max.
Al	Balance	
Mg	2.5	4.2
Zr	0.6	1.8
Mn	0.1	1
Fe	0	1
Si	0	1
Ti	0	1

Powder Particle Size

Generic Particle Size Distribution	20 - 63 µm



SEM micrograph of EOS Aluminium Al5X1 powder

HEAT TREATMENT

Description

Heat treatment procedure

Steps

Direct ageing heat treatment

6 hours at 400 °C. Immediate gas quenching (air cooling with maximal air flow).

Preferred inert atmosphere during furnace treatment.

PROCESS DATA SHEET

EOS Aluminium Al5X1 for EOS M 290 | 40 µm

EOS M 290 - 40 µm - TRL 3

System Setup	EOS M 290
EOS Material set	Al5X1_040_CoreM291_100
Software Requirements	EOSPRINT 2.11 or newer EOSYSTEM 2.15 or newer
Recoater Blade	HSS (High Speed Steel)
Nozzle	EOS Grid Nozzle
Inert gas	Argon
Sieve	75 µm

Additional Information	
Layer Thickness	40 µm
Volume Rate	4.8 mm^3/s

Chemical and Physical Properties of Parts

The chemical properties of the parts are the same as that of the powder.

Microstructure of the Produced Parts

Defects	Thickness	Result	Number of Samples
Average Defect Percentage	40 µm	0.15 %	-

Density EN ISO 3369	Thickness	Result	Number of Samples
Average Density	40 µm	2.69 g/cm³	-

Mechanical Properties

Mechanical Properties Heat Treated

ASTM E8 Room Temperature | 40 µm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	380	415	14.5	-	-	-
Horizontal	385	415	14.5	-	-	-

ASTM E8, strain rate 0.00762 mm/s

Mechanical Properties As Manufactured

Room Temperature | 40 µm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	240	310	23	-	-	-
Horizontal	260	310	23	-	-	-

Hardness

Heat Treated	
Value	127
Unit	HBW 2.5/62.5

Heat Treated	
Value	71
Unit	HRB

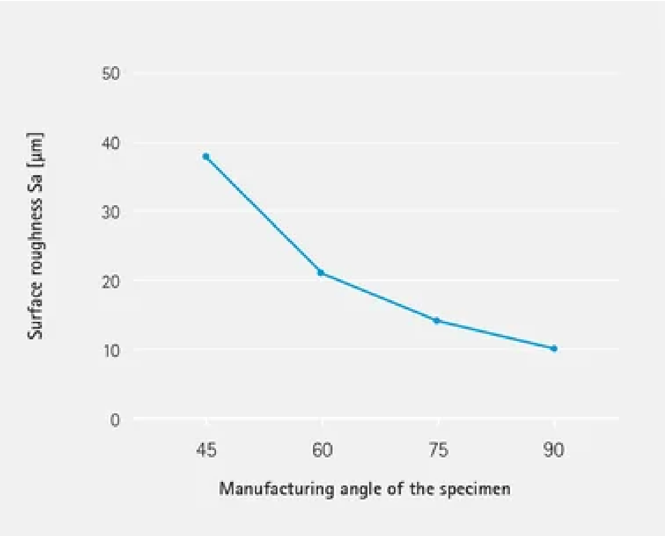
As Manufactured	
Value	86
Unit	HBW 2.5/62.5

As Manufactured	
Value	40
Unit	HRB

Thermal Conductivity

ASTM E1461-13	Orientation	[W/m*K]
Heat Treated	Vertical	132

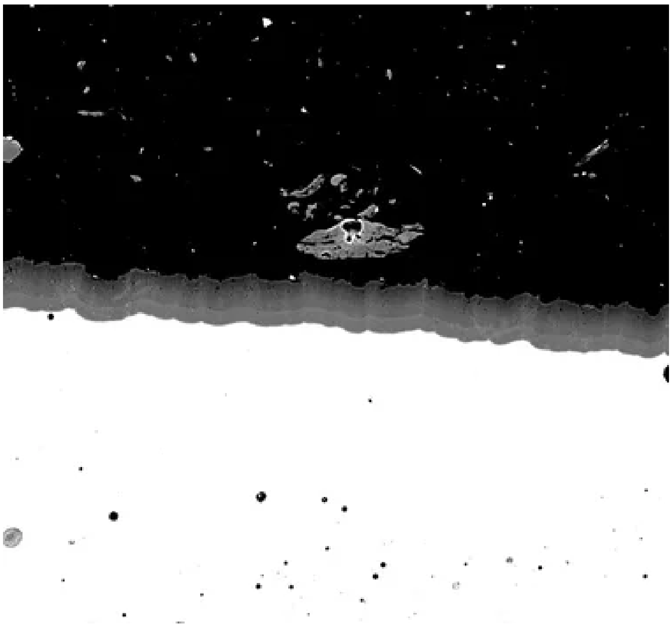
Surface Roughness



Specific heat capacity at 25°C

AST E1269-11 Heat Treated	
Value	0.87
Unit	J/g/°C

Anodization



SEM image of the anodizing layer.
Anodized according to Mil-A-8625 Type III

Thickness of anodization layer depends on used anodization process.

Electrical Conductivity

ASTM E1004-17	Orientation	Typical Electrical Conductivity [%IACS]
As Manufactured | 40 µm |	Vertical	23

ASTM E1004-17	Orientation	Typical Electrical Conductivity [%IACS]
Heat Treated | 40 µm |	Vertical	34

HEADQUARTERS

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Status as of 18.08.2026. Subject to technical modifications. EOS is certified according to ISO 9001.